Fat-Adapted Exercise:

Most well-trained endurance athletes can run at speeds approaching their anaerobic threshold for nearly two hours, after which the body's glycogen stores get rapidly depleted, unless the athlete has run many hundreds of kilometres in training at lower aerobic intensities, often with continuous steady state aerobic runs of well over two hours, which encourages preferential oxidation of fats for fuel at higher sustained work rates and power outputs.

Major Problem Number One: **The tiny fuel tank**
The limiting factor for all endurance athletes, no matter which sport, is the relatively tiny fuel tank of glycogen that starts to sputter on *empty* at around that two-hour mark, creating the legendary *wall* that afflicts marathon runners, cyclists, endurance kayakers, multi-sports athletes and triathletes alike.

Arthur Lydiard's training-based Solution:
Arthur Lydiard's famous sustained long runs, performed without supplementing carbohydrates past two hours, enabled athletes to eventually adapt to using fats for fuel at reasonably high percentages of their maximal aerobic running capacities. As the athlete improves in his or her ability to run long distances at comfortable aerobic efforts, several changes occur in the physiology:

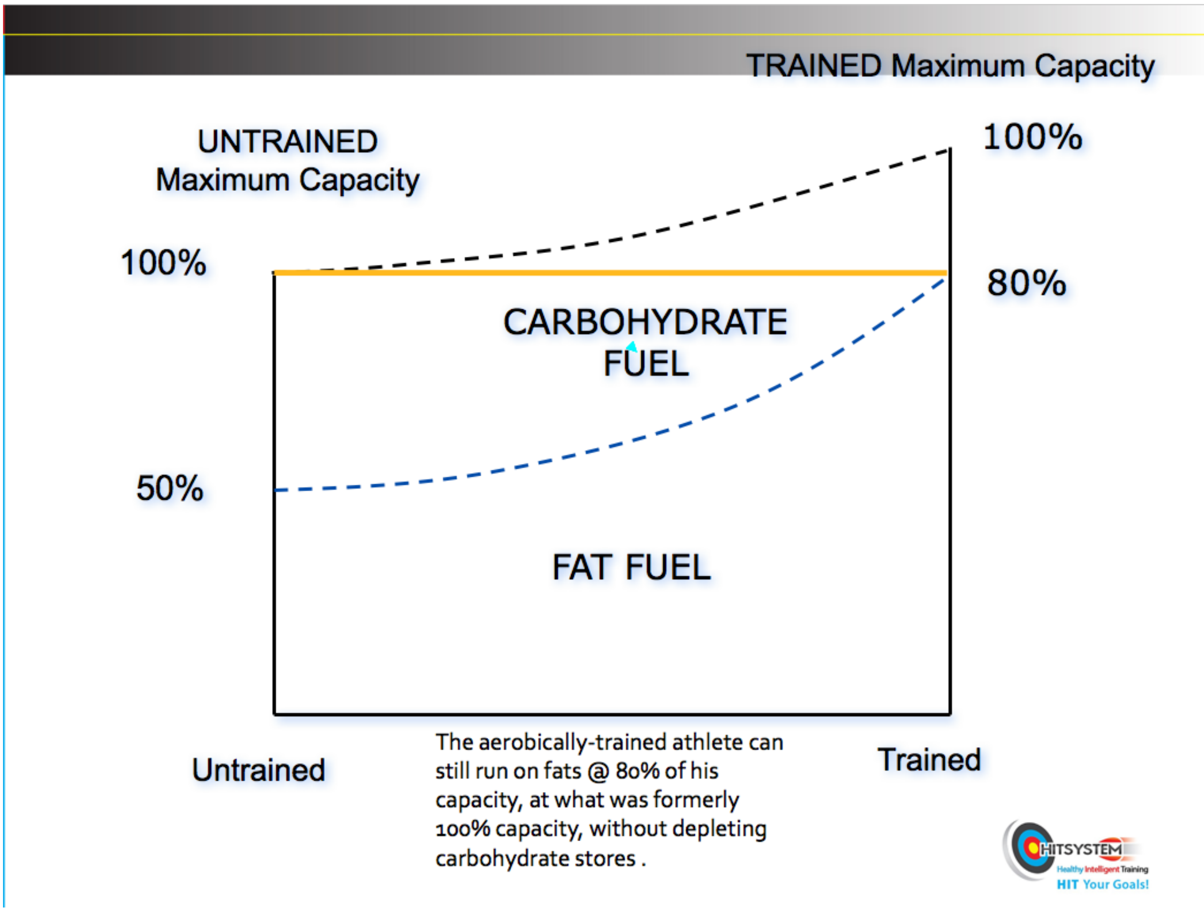
1. Sustained low aerobic pressure on the capillary system invokes *angiogenesis*, or the formation of new capillary networks throughout the muscles being exercised; this includes the heart muscle as well as the running muscles; both circulatory systems benefit.
2. Because more blood is able to be supplied at a lower pressure to the working muscle, exercise becomes easier. The extra capillary networks are able to supply fuel to the muscles quicker, and carry away waste products and metabolites like carbon dioxide and lactic acid more quickly, too.
3. The athlete eventually runs more and more on fats as a primary fuel source, instead of on *the tiny fuel tank* of stored carbohydrates (glycogen).
4. Eventually with the prolonged endurance runs, the athlete will reach a state of near-tirelessness at all aerobic speeds below his or her anaerobic threshold, as the body can store many hours' worth of energy in its fat depots. As long as the athlete stays hydrated and the electrolyte levels are maintained, aerobic exercise without true fatigue should be able to continue for hours utilizing fatty acids.

The Holy Grail of Lydiard endurance base training is the adaptation to burning fatty acids as preferred fuel instead of stored carbohydrate. To date, orthodox sports nutrition has not been able to find a way to get around the *tiny fuel-tank model* for endurance athletes. However, what any sports scientist or nutritionist will know is that we can go a very long way on fat stores, as long as our sustained work rate is at lower aerobic intensities which allow fatty acid metabolism to power us along.

Conventional wisdom in sports nutrition tells us that it is impossible to run at maximal aerobic power on fats alone. There are many studies that demonstrate the relative inefficiency of burning fat for fuel compared with burning glycogen.

Unfortunately, most studies were done with athletes who trained and ate conventionally. (They ate lots of carbohydrates before training and competition).

Those athletes who trained extensively with long steady aerobic efforts were able to gradually transition from burning about 50% sugar/ 50% fat at steady aerobic running rates to burning about 20% sugar/80% fat at aerobic steady state intensities, as this diagram illustrates. The formerly *untrained* 100% Max VO2 work rate becomes a fat-burning 80% of the newly achieved *trained* 100% Max VO2, which is about 125% of its former level.

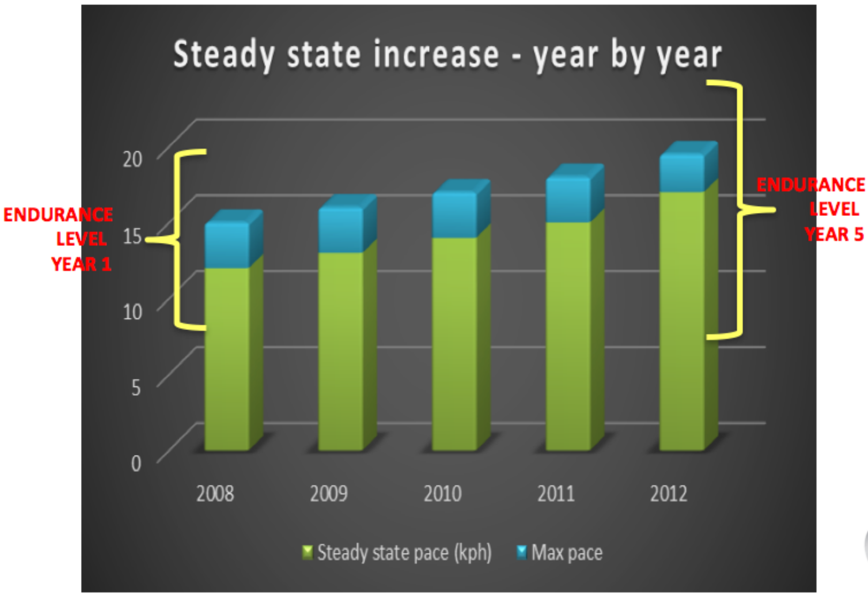


From *Distance Running*, TAFNEWS, Dr David Costill, 1979.

The diagram below is based on real data from athletics coach René Borg in Ireland. Notice how the same progression with increased oxygen uptake is obvious, with training loads that were initially above the maximal oxygen uptake becoming easily achievable and *aerobic* after 5 years of constant aerobic endurance running.

ACTUAL AEROBIC PROGRESSION

Case study: René Borg (Dublin)



However, even for athletes who gradually increased their aerobic training loads the Lydiard way, the *tiny fuel tank* still limited sustainable high-aerobic work rates because the most that the body seemed to be able to sustain in repeated laboratory testing was about 80% of maximal intensity, utilizing fats, which is still a few percentage points slower than the glycogen-burning precipice of the high-power-output *anaerobic threshold*.

Until recently, every sports scientist worth his salt could tell you that intensities at the anaerobic threshold were only achievable with glycogen-burning, and that fat simply couldn't provide enough power output to sustain maximal oxygen uptake. *Until recently,* only one published study had ever bothered to look at athletes who had become accustomed to *higher fat intakes in their diets before* they were tested on their capacity to burn fats in the lab.

The FASTER Study recruited a group of carefully-screened elite-level ultra-runners from two distinct dietary camps. One group, the Low Carb Fat-Adapted, consumed 10-12 % of their daily calories from carbohydrates and 70 % from fat. The athletes in this group were chosen for having followed a sustained regimen of low-carb eating prior to the study. Individuals from this group were paired with runners of equal ability and body composition in a High Carbohydrate Conventional group. This other group ate a more typical endurance athlete diet consisting of 60 % carbs and only 25% fat.

Prior to this study, conventional science said the absolute maximum amount of fat an athlete can burn is upwards of 1.0 grams/minute, with most highly trained athletes falling into the 0-45-0.75 grams/minute range [(1)].

The FASTER study published in 2015 (Fat-Adapted-Substrate oxidation in-Trained-Elite-Runners) **shattered accepted limits** with a *mean* fat oxidation rate amongst fat-adapted athletes of 1.54 grams of fat per minute and a maximum rate in one athlete of 1.8 grams of fat per minute, which was nearly *twice* the rate formerly believed to be the *absolute human limit*. The *lowest* fat oxidation rate recorded in the fat-adapted athletes was 1.1 gms of fat per minute, which is *still 10% higher than the formerly recognized absolute human limit.*

1.Venables et.al.; *Determinants of fat oxidation during exercise in healthy men and women: a cross-sectional study, Journal Applied Physiology 98;160-167,2005.*

To understand the differing types of metabolism regarding carbohydrate oxidation and fat oxidation, we need a measure called RQ (Respiratory Quotient). This is a simple calculation in a laboratory setting of the ratio of carbon dioxide exhaled to oxygen inhaled. A ratio of 1:00 indicates 100% carbohydrate metabolism for the exercise intensity being evaluated. In other words, an athlete with an RQ of 1:00 or close is a sugar-burner, ripping through muscle and liver glycogen at intensities as low as 35-50% of VO2 max. By contrast, an athlete who metabolizes fats efficiently will have an RQ of about 0.7, and will be burning mostly fats at higher aerobic intensities, conserving the higher-octane glucose stores (glycogen) for the business end of an event or extended duration training session when required.

As far as sustainable power outputs go with fat-adapted athletes, there are three interesting case studies discussed in *Mark Sisson's* excellent book, *Primal Endurance,* that are worth *directly citing*. The first notable case study was with American 100-mile record-holder, Zach Bitter, who was laboratory-tested in a treadmill test run at 7 minutes per mile, with a maximal fat-burning percentage of 98%, while running at 75% of his Max VO2.

Another interesting case study was with Dr Peter Attia, a San Diego physician and *deep experimenter*. Before Attia transitioned to a fat-adapted (ketogenic) eating pattern, he burned 95% carbohydrate and only 5% fat at an all-day comfortable pace on a treadmill test. At a heart rate of 104 beats per minute (very slow pace) he hit a point of burning half carbs and half fat. After a strict 12-week period of eating *ketogenically*, his repeat lab test results were *nothing short of astounding.* At the all-day pace, Attia burned 22 percent glycogen and 78 percent fat! The heart rate where he hit the 50/50 ratio of carbs and fat increased to 162 beats per minute–hammering pace!

When Attia bumped his pace up to the red-line anaerobic threshold (AT), he burned 100 percent glycogen before, but 70 percent glycogen and 30 percent fat at AT during ketogenic eating. Furthermore, Attia reached AT at a higher wattage as a *ketogenic athlete*. This caused him to speculate that ketones may blunt the effects of acidosis on the bloodstream, enabling improved performance not only at comfortable fat-burning intensity, but also at red-line pace. Interestingly, when Attia completed his test by revving up to all-out VO2 Max pace, he experienced a slight decline in performance while ketogenic. *His carefully recorded data suggests that with regard to fat-adapted ketogenic eating, there is a substantial benefit at virtually all intensity levels short of absolute maximum effort.*

The third case study was world champion amateur 70.3 triathlete Sami Inkinen, who has shown with precise laboratory data that ultra-low carb eating (less than 10% of total calories) eating patterns can dramatically improve endurance performance. In only three months of a high-fat, low-carb eating pattern, Inkinen went from an extreme sugar-burner (around 95% of total fuel usage at a relatively high intensity 300 watts) to a remarkably fat-adapted endurance machine. At that same 300 watts of energy output after his dietary modification, Sami transformed his substrate utilization from 95% glucose to around a 50/50 split between glucose and fat, and doubled his fat oxidation to 400 calories per hour. This was at a laboratory-confirmed 300 watts energy output rate equating to peddling around 25 mph (40+ kmh!) on flat ground!

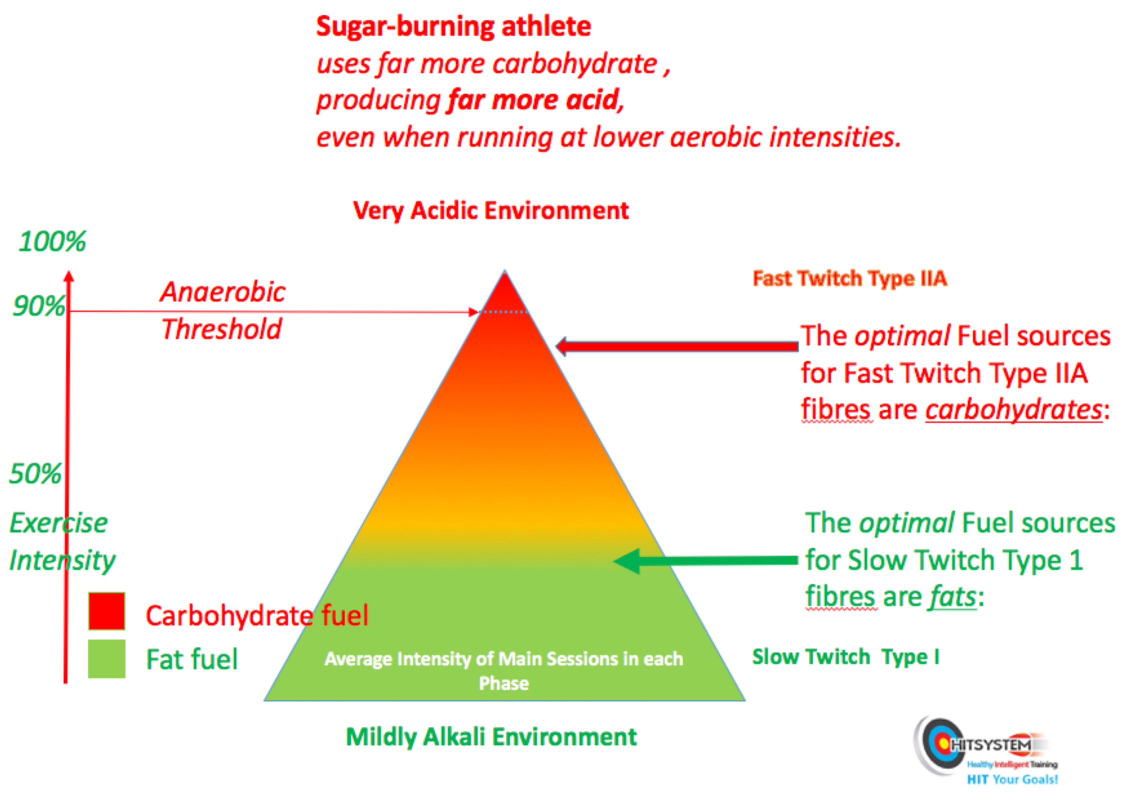
Even more amazing was Inkinen's "time to bonk" (zero blood glucose) calculation at lower intensity, extrapolated from substrate utilization data at his three performance tests. The calculation predicts when the subject would run out of energy, assuming no additional ingested calories. At his first test as a sugar-burner, he calculated a bonk time of 5.6 hours at the low intensity of 200 watts output. At his 2014 test, after five years of fat-adapted eating and training, the calculated *time to bonk* at 200 watts increased to 87 hours!

Sugar Burners and Fat burners.

It is sometimes very surprising to find out who is a sugar-burner and who isn't. Even professed hard-core endurance sports exponents at the elite level, who have been training over long distances for years, can be big-time sugar-burners.

A man I have advised since 1987, John Meagher, who is now 55 years old, had been one of Australia's best-performed age-group runners and triathletes for many years, after running a debut marathon of 2hr 16 minutes in 1989, running or cycling high volumes of aerobic training for many years. Earlier this year he explained to me that he often felt tired straight after work (he's a teacher): he was *too tired* to run just a few kilometres with the boys he coached for the school cross-country squad, after school. He complained of being light-headed and dizzy on long hilly runs, and not enjoying his long runs any more. He was functioning as a sugar-burner, and possibly was hovering on the edge of insulin resistance, or early type two diabetes. He needed to become fat-adapted again, through diet.

The diagram below perhaps illustrates John's metabolism when he was a sugar-burning athlete.



I noticed at a few family meals that John ate masses of carbohydrate foods like bread and toast, with the regulation cornflakes for breakfast. If there was salad with lunch or the evening meal, it was minimal, and only a nod in the right direction. I told him about the *fat-adapted* diet that *Primal Endurance* author *Mark Sisson* advocated, that featured an inclusive spread of as many of the coloured vegetables as possible, often simmered with butter to allow for release of the fat-soluble vitamins A,K, D and E, and minerals like magnesium, in the plant foods. I recommended a regular, filling breakfast with eggs and organic non-nitrite bacon, and left the rest to him.

He also dosed up on a few tablespoons of coconut oil each day; sometimes in a cup of tea. Coconut oil is very rich in medium chain triglycerides which are healthy saturated fats that get quickly converted in the liver to ketones, which are fat-like molecules which are also water-soluble. While the fat-adapted athlete's muscular system runs off fats by the process of *lipolysis,* the brain can utilize ketones as a rich fuel source in the absence of rapidly dwindling glucose. *Until recently*, it was thought that the brain could only derive energy from glucose, however recent science has shown that the brain can derive up to 15% more energy from ketones than from glucose. Brain energy is very important at the business end of an endurance event, and it appears that the amazing body preferentially diverts ketones to the brain for brain energy and fatty acids to the working muscles, in *fat-adapted* athletes.

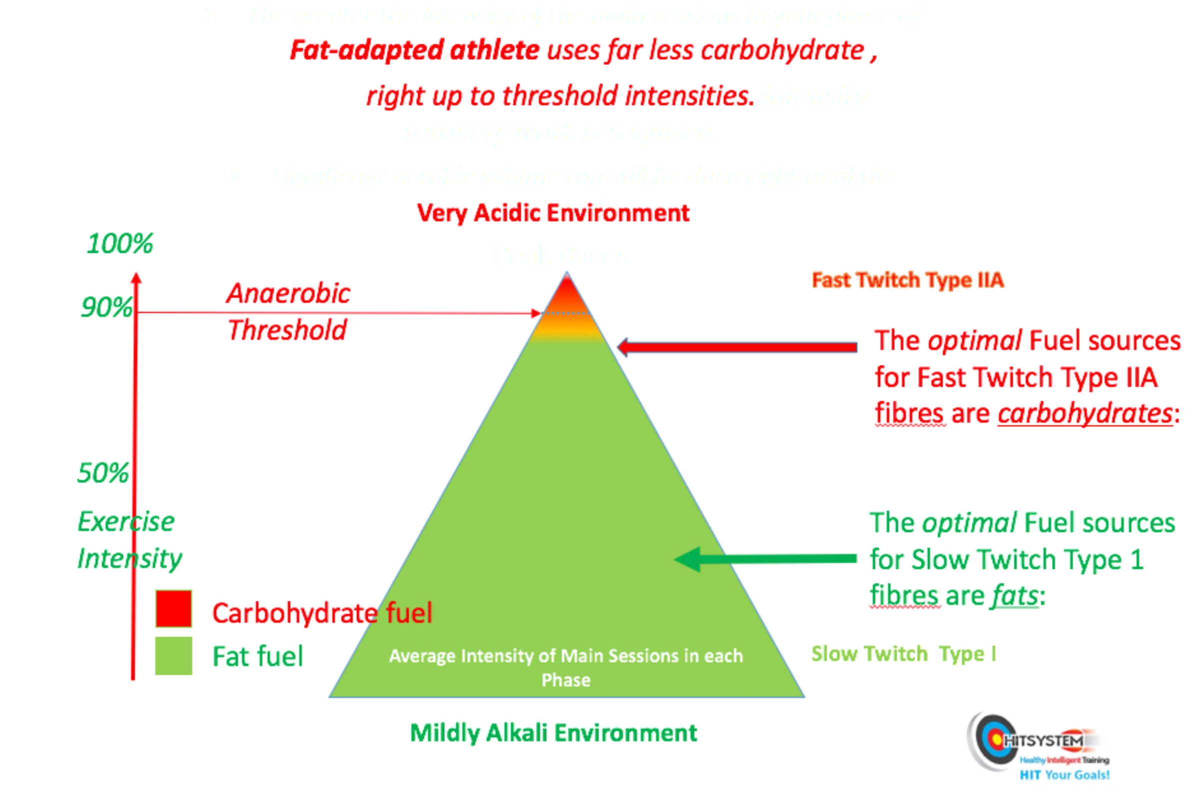
Fast forward a few months, and when I last spoke to John, he reported that he was feeling the best he had for years, able to run over 40 kilometres on big hills again, without feeling fatigued. He never had pangs of hunger any more, and his energy levels were always reliable and constant. He is now looking forward to running his best marathon time for several years, without fear of *hitting the wall.*

John broke the Australian M55+ best marathon time by 11 seconds on October 14th, in the Melbourne marathon, running 2 hours, 37 minutes, and 16 seconds, feeling strong and full of running, with a hard last kilometer in warm, windy conditions.

He recovered very quickly from the marathon, and may attempt to lower his time further in the 2019 Tokyo marathon, on March 3rd.

The diagram below illustrates John's fat-adapted metabolism now.

The Fat-Adapted Athlete



One strategy for race day that we discussed was for John to eat a high-carbohydrate, high-fat meal the night before his marathon (a baked potato full of cream cheese or sour cream fits the bill), and to keep very slow, fat-burning aerobic intensities of running going right up to race eve, cutting back the duration of extremely long weekend runs by 20% in each of the two weeks leading into his race. We'll see how he goes.

It takes about three weeks for a sugar-burning athlete to get fat-adapted by gradually increasing the amount of fat in his or her diet, and cutting carbohydrate consumption right back. There are no downsides, with a fully inclusive diet rich in all vegetables, some meat, and saturated fats, which are totally necessary for the full absorption of fat-soluble nutrients and minerals. You just cut out sugary foods and white flour pastry goods, and enjoy the rest. You can eat as much in the way of vegetables as you like, and you'll stay lean, especially if the vegetables are cooked with saturated fats like butter, which liberate the fat-soluble vitamins A, K, D and E, as well as minerals like magnesium, from orange, red or dark vegetables, or green leafy vegetables like kale or spinach or brassicas like broccoli and Brussels sprouts.

A big caveat here is that we must avoid certain oils in our diet if we wish to maintain our cellular machinery at the highest level. These are the processed polyunsaturated seed oils that are in so much processed food these days, like cottonseed oil, used to deep-fry your fish and chips, or the Canola oil used in your margarine. If you have too many of the wrong oils, your metabolism can be substantially altered for the worse.

There are Oils… and then there are Other Oils

Omega 3 oils and Omega 6 oils are both necessary for human health. They should IDEALLY be ingested in even quantities, or a ratio of 1:1. Omega 6 oils help mediate the immune response. However, if the ratio of omega 6 oils to omega 3 oils increases a great deal past the 'IDEAL' 1:1 ratio, there is an inflammatory cascade in the body that is thought to predispose to many of the degenerative diseases and cancers that the Western World suffers from.

Ischaemic heart disease was virtually unknown until earlier last century when American-grown vegetable and seed oils high in omega-6 fatty acids flooded the markets, after the traditional supply of coconut oil and palm oil from the Western Pacific and South East Asia was disrupted by the Japanese during World War Two. Some grain or seed oils were hydrogenated to create margarine.

Margarine's biologically useless trans-fats and omega 6 oils compete with essential omega 3 fatty acids for receptor sites in our brain and nervous system, and can muck things right up hormonally and in the expression of gene pathways in cells. These are the nasty fats in your favourite packet of salt and vinegar potato crisps, unfortunately, or in your battered fish with chips. If I ever munch on some, (they are, unfortunately, *very* tasty!) I immediately dose up on fish oil capsules, high in omega-3 fatty acids, to redress the ratio.

The Heuristic *Theory of Everything*

At a seminar in Melbourne in 2016, Dr Dan Murphy, a brilliant chiropractic research presenter, and a popular professor at Life Chiropractic College West in San Francisco, outlined his method of keeping the wholism of good nutrition at the forefront with his undergraduate students, with what he termed his *Heuristic Theory of Everything* teaching method. (*heuristic | adjective: enabling a person to discover or learn something for themselves: a *hands-on* or *interactive heuristic approach to learning*.)

Instead of talking about these nutritional theories, he gets his students to assay changes in their lipid profiles with pin-prick sampling of their blood every couple of weeks, over several months. They experience for themselves how different ratios of omega 3 to omega 6 fats affect their health and energy status, and by gradually implementing strategies to normalize those levels, they *live* the lesson.

The SAD Diet

The *Standard American Diet* and its copycat, the *Standard Australian Diet*, are way out of control as far as the ideal ratios for good health are concerned. Dr Murphy expressed his shock that in the cohort of health-conscious undergraduate health practitioners he was teaching, one female participant's levels had *blown out* to as much as 1:70, and several more students were at the levels that are typical of many modern Americans; around 1:15 to 1:30! Dr Murphy mentioned that most of the cohort *appeared* to be fit and healthy, even if their ratios vastly exceeded the *healthy* range.

Who would know what *irreversible* processes had been started, especially in the girl who had the 1:70 ratio?

Finding how otherwise *normal* American kids could have such terrible levels of omega 6 in their bodies is not hard; *vegetable oils* are in virtually every packaged food product.

Often, they're not even oils sourced from *vegetables*, but processed from *seeds* that would otherwise be *left-overs* after processing in commercial food production. Why chuck something out when you can make a profit from it? Unfortunately, these oils are often heated and reheated to levels of rancidity in fast food outlets and general food manufacture.

Elevated levels of vegetable oil omega 6 fats in the body cause an elevated inflammatory response, and current scientific thinking implicates *inflammation* as a root cause of many degenerative conditions.

I find it useful to picture the two necessary types of essential fatty acids, omega 3 and omega 6, as entering two parallel and inter-related metabolic cycles, each with its own desirable outcome. If there is an imbalance in the *desired* 1:1 ratio, (mostly with an excess of omega 6 fatty acids due to the prevalence of vegetable and seed oils in the SAD diet) then more and more omega 6 will pile into the omega 3 cycle, reducing the metabolic benefits in proportion to the omega 3: omega 6 ratio, and increasing the inflammatory cascade similarly, in inverse proportion.

C-REACTIVE PROTEIN

Is an enzyme produced by the liver in the presence of acute inflammation that can occur anywhere in the body, however it is also highly correlated with risk of atherosclerotic heart disease. It is a good idea to keep your levels of C-reactive protein down as far as possible. The most accurate test for C-reactive protein levels is the HS-CRP test (High sensitivity C-Reactive Protein test).

Keep those Cholesterol Levels Up

Some of what I'll share here could be quite alarming, but it's very well backed-up by recent research. In my personal research of the *cholesterol myths* over the years, I'm at the stage now where I don't worry one iota about cholesterol; what I do try to control is lifestyle stress and inflammation or oxidation of fatty acids in the body. Inflammatory hot-beds in the lining of arteries tend to oxidize circulating lipids into calcific atherogenic plaques. Elevated saturated fat intake has *not* been demonstrated to have an association with heart disease, despite every attempt by major vested-interest groups to demonize what was always part of a healthy diet until the 1950s.

What if the total cholesterol in my bloodstream is a bit higher than currently suggested good levels for health? That's fine, my brain absolutely needs it to function at 100%. Every cell membrane in the body incorporates cholesterol, including the brain.

In my extended family, we witnessed first-hand the harmful effects of cholesterol-lowering medication on my wife's mother, *Lalli*, who was turned from a healthy, active woman in her early 70s, who exercised in dance classes with her friends several times a week, to someone so weak she could not climb the steps in her pantry or even hold her hair-dryer up for more than a minute or so. As a very healthy and high-functioning post-polio survivor, she should never have been placed on cholesterol-lowering medication, however, for some reason she was seen by a lady doctor who routinely prescribed her a statin drug, *Lipitor*.

When *Lalli* started to feel unwell and *out of sorts* on her medication, she advised her doctor, whose response was "Oh! Well, if you don't lower your cholesterol, you won't live very long then!"

It took many months of supplementation with Coenzyme Q10, the mitochondrial enzyme that statin drugs are notorious for depleting, and chiropractic adjustments to re-stimulate her central nervous system, to get *Lalli* back on track.

What about the *clogging of arteries* in the heart? Yes… what about that emotively-termed myth? Ingestion of saturated fats such as those found in organic butter and grass-fed beef *normalizes* blood lipid profiles to healthy levels; saturated fats *haven't* been proven to raise risk of heart disease at all.

Eating simple carbohydrates and grain cereals, however, *will* create the inflammation necessary to drive up low density lipids and the formation of atherosclerotic plaques in coronary arteries, *not* eating fats! Stay off grain cereals and *white anything* (rice/ bread/ sugar) and your inflammatory load will lighten straight away.

Getting off simple carbohydrates and elevating your omega 3 fish oil intake will lower your triglyceride levels nicely. To raise your HDL (High Density Lipid) to desired levels you need to exercise steadily on several days a week, for periods of at least 30 minutes. Walking *ticks the box* there nicely– preferably in a park or bushland setting, getting in contact with nature.

If you can also walk barefoot for half an hour or so, that could be very good, as there is a new body of research into the positive effects of *grounding* or *earthing* the body with the constant electron-rich surface of our planet. You literally *plug into* the biggest source of antioxidant electrons available. Direct contact with the earth's field of electrons can explain why we usually feel so enervated after walking on the beach and diving into the surf. By the seaside, you are benefitting from masses of *negative ions* generated by the action of the wind on the surf, and by diving into the surf, one is literally immersed in a *sea of electrons*. A similar effect is reported near waterfalls and running rivers.

Earthing mats that hook into the *earth* wires of normal homes have been available online for several years. It is possible to sleep overnight with one's bare feet resting on cotton pads with highly conductive micro-thin silver strands woven in. Because no other current is involved, it is totally safe. The same effect could be obtained by connecting any type of conductive foil to a wire that is attached to a tent peg in the garden. (No! Not a plastic tent-peg!).

You could simply sit with bare feet on your back lawn for half an hour to receive the suggested benefits, while getting a bit of sun on your arms and legs; the surest way to stimulate your body's own production of vitamin D3.

When Is a Vitamin No Longer *just* a Vitamin?

Vitamin D3 is a cholesterol-based steroid hormone that until recent years was relegated to the sidelines as a vitamin involved in the normal clotting of blood, and in the prevention of the bone-softening disease, rickets. However, in the last two decades much research has come to the fore that suggests that vitamin D deficiency is extremely common, even in active people who spend a lot of time outdoors.

I considerably annoyed my naturopath early on with my *little issue* when I queried why I needed to have my blood levels of vitamin D tested when I was outside in the sun on a mountain bike most days. It turned out that she knew far more than me about the subject, and in the 20 years since I had studied clinical nutrition, vitamin D had been moved to the front of the queue in terms of *importance*, and was now described as a valid steroid hormone. It had acquired a dirty big US Government-funded Not-for-Profit Group, the *Vitamin D Council*, such was its rapid elevation.

Guess which vital substance is depleted by the anti-seizure medication I had been on for a decade, to prevent brain-tumour-related seizures, before it was changed to a much-better medication, this year? Phenytoin, or *Dilantin*, induces deficiency in calcium, and vitamin B12, as well as vitamin D. For the last thirty years I have usually eaten several free-range eggs a day, lots of baked or roasted coloured vegetables, lots of cheese, and lots of fresh green salads.

Fresh lightly-braised salmon is a favourite, along with mashed avocado, feta cheese, and broccoli. Himalayan rock salt, loaded with over 80 trace minerals, is superb on eggs, along with black pepper. Eggs are rich in Vitamin D, and are good sources of Vitamin B12, Omega-3 fatty acids, calcium, and zinc, as well as other vitamins, minerals, and carotenoids (precursors of vitamin A). The egg is a powerhouse of disease-fighting nutrients like lutein and zeaxanthin as well.

While vitamin K1 is preferentially used by the liver to activate blood clotting proteins, vitamin K2 is preferentially used by other tissues to *deposit calcium* in *appropriate* locations, such as in the bones and teeth, and *prevent* it from depositing in locations where it does not belong, such as in the soft tissues and arterial walls.

Fresh dark leafy green salad leaves are a rich source of vitamin K1. These are also sources of phytic acid, which actively inhibits absorption of minerals like iron and magnesium, so it’s best to break down the phytates by cooking or simmering on a low heat. To access the fat-soluble vitamins, which include vitamins A, K, D, and E, it is best to cook with a healthy source of saturated fat like *grass-fed* butter or cold-pressed olive or coconut oil. Amongst the dark leafy green vegetables, the one with the highest source of vitamin K1 is *kale*, a form of spinach.

The richest sources of vitamin K2 are in goose liver pâté or traditionally fermented cheeses. The delicatessen or whole food store is the best place to source your vitamin K2 naturally. Since researching this, my opinion of people who regularly fare on pâté and cheese has increased hugely. At one time years ago, a potential girlfriend was wooed away by a suitor who invited her to have some pâté at a café, earning himself the jealous moniker ‘Pâté’ from me. He not only got the girl, but he probably has great bone structure and cardiac health now, too.

Getting off the Anti-Seizure Drugs with Nutrients

Unfortunately, I am currently on these drugs to prevent seizures with *100% certainty*. I hate the idea of being on any drugs, and my aim eventually is to reach healthy, fully-functional advanced age without being on any medications whatsoever. To ensure this, I have to make sure that I don't have any more major seizures, as well. This is entirely possible if I remove myself from stressful situations and stick to the proven drug regime for now. It's a matter of submerging my ambition to be totally drug-free for a few more years, while at the same time propping up my neurological health with an abundance of neuro-supportive nutrient-dense foods and minerals.

I definitely never have seizures if I eat and sleep well, go for my daily walks, and take my anti-seizure medication every morning and evening. However, I have found myself hours from home, having forgotten my medication on several occasions, with no access to my prescription, and I got around OK for a few days on large amounts of fish oil and magnesium supplementation from the supermarket, with no sign of a seizure. This is promising.

With another year totally free from seizures and their neuro-excitatory damage, while I build up my neuronal mass with the nutritional and lifestyle strategies I described earlier in this chapter, I believe it's possible to slowly wean myself off anti-seizure medication and replace it with omega-3 fish oil, magnesium, coconut oil, and the amino-acid L-carnosine. These have all shown to be neuro-protective and seizure-preventive to varying extents.

I read recently that if one limits one's carbohydrate intake per day to 25gms or less, then the blood glucose levels will be far too low to fire a seizure. *Even though blood glucose levels will be on the lower side, overall brain energy and mitochondrial energy output can still be maintained at a high level with ketone bodies*. When the body has adapted to fats and is manufacturing sufficient ketones, this is known as *ketosis*, or the *ketogenic state.*

A great way to raise your beneficial HDL levels is to use coconut oil regularly as a condiment. Coconut oil is anti-inflammatory by nature, and is rich in lauric acid, a medium-chain triglyceride fatty acid (MCT) that is particularly amenable to human metabolism. MCTs get rapidly metabolized by the liver into water-soluble variants of fatty acids called ketones which as well as giving richer energy yields than glucose, have the benefits of far less free radical formation within the *energy furnace* cells of the body, the mitochondria.

If one induces ketosis with the ketogenic low carbohydrate/higher fat diet, then the brain can still get its energy requirements from the ketones, once the system has become fat-adapted. If you're still in the process of getting fat-adapted, ketone production will be sluggish and intermittent. Getting fat-adapted with your diet takes about three weeks, whereby one gradually introduces higher-fat foods while progressively limiting carbohydrates.

Ketosis is not to be confused with *keto-acidosis*, which is a symptom of diabetes and renal failure, where the breath smells of acetone. Medical doctors, who don't study nutrition, may confuse ketosis with *keto-acidosis,* which could explain why healthy fat-adapted eating is seen as a fad.

During the fat-adaptation phase, while you are aiming to be in ketosis, you can supplement with coconut oil, which will ensure a good level of ketones is available for brain energy.100ml of coconut oil, or a couple of tablespoons that can be spread over a salad or mixed in with your meal, will do the job. If you haven't had much coconut oil before, you may find that it is an excellent laxative; just a warning before you put on your tight-waisted white bell-bottomed stretch-polyester trousers for a night at the disco.
I don't envisage a calcium or vitamin D3 deficiency, or a K2 deficiency from the years of anti-seizure medication I have had already, as I eat such large amounts of all the right foods.

With neurodegenerative conditions, which are often coined *Type III Diabetes*, or *Brain Diabetes* there is insulin resistance in the brain, thereby starving the brain of its glucose uptake. *However*, it has been shown that the brain can metabolize ketones derived from the MCTs in coconut oil for energy, bypassing the need for insulin uptake. Not only that, the energy yield from ketones is higher than for glucose, and it is likely that the metabolism of ketones within the mitochondria does not produce the same degree of harmful Random Oxygen Species as glucose metabolism would. In essence, there is a net anti-inflammatory effect from ketone metabolism as compared to glucose metabolism, and ketones are said to be very *clean* fuels for the body.

It could well be that the use of coconut oil early in the treatment of brain tumours could feed the healthy brain cells while depriving the tumour cells of the only source of rapid energy they can process: glucose. Certainly there is some evidence that coconut oil can induce *apoptosis* or cell death in bowel cancer.

What interests me personally is that this ketogenic approach to diet has been successfully used for over 90 years in the treatment of epilepsy and type two diabetes, and has been shown to be very effective in preventing seizures in epileptics. However, it is not in the interests of the large carbohydrate producers (*Big Grain*) who dictate a great deal of nutritional science these days.

Nutritional science is as bad as any other science in terms of being personality-driven, industry-driven and populist-driven. The power of *orthodox* nutrition to skew us away from our best health outcomes is exemplified by the recent very public trial of the much-published Dr Tim Noakes, in South Africa. Dr Noakes *was* the doyen of carbohydrate proponents, and a world leader in research into carbohydrate use in endurance sports.

Dr Noakes started to suffer Type II diabetes complications a few years ago. His father and a brother had succumbed to the condition previously, so he decided to embark on a ketogenic approach to his personal nutrition and lost 44 pounds (20 kg) in the process, reversing his Type II Diabetes, while returning to good enough shape to run again. He then *back-flipped* from his carbohydrate diet tenets established over a lifetime career in research, and went in another direction altogether.

The backlash to his newly affirmed dietary position was vicious, with high-profile *colleagues* and former acolytes in the nutritional research world going as far as saying he'd "lost his mind". He was reported to the health practitioners' registration board in South Africa by a hospital dietician who got very upset with a non-dietician advising people on nutrition apparently. He won his case at the highest level in the court system in South Africa, however the remarks on his case were that his views were *not orthodox*, or *not in keeping with the majority viewpoint*.

It could be many years before the *majority viewpoint* is allowed to *catch up with the times* without getting practitioners deregistered.

There is obviously a great deal more to the reason why divergence from the accepted norms of the SAD diet will invoke the wrath of registration bodies. Apart from the obvious challenge to the existing nutritional paradigms, upon which many academic reputations and huge corporate fortunes have been made, I'd say that large vested-interest groups in the food industry have a big say *behind the scenes*, just as the pharmaceutical giants who manufacture statin drugs and sponsor many medical conferences and University Chairs have a large say in the amazing denial of recent science that debunks their views. It's all at the expense of the public's long-term physical and cognitive health.

To me, it is absolutely unacceptable that publicly-funded agencies like the Heart Foundations continue to promote an outdated paradigm like the lipid hypothesis of heart disease. It is *outrageous* that the paradigm is still believed, over 20 years after having been debunked in peer-reviewed studies, and it is even more outrageous that the statin drugs are still on the market.

If It Walks Like a Duck and Talks Like a Duck..

It probably is. The very nutrients which have been demonstrated by peer-reviewed research to have a positive effect on reversing atherogenic plaques are criticized in carefully-placed off-hand comments by the American Heart Association. Red palm oil is one of the richest sources of two of the most powerful forms of vitamin-E and vitamin-A antioxidants in nature: the tocotrienols and the carotenoids.

Tucked away on one of the American Heart Association web pages under the helpful title *About Cholesterol* is the following comment, handily placed to throw just enough caution into the mix about these healthy non-drug alternatives:

Some tropical oils, such as palm oil, palm kernel oil and coconut oil, also can trigger your liver to make more cholesterol. These oils are often found in baked goods.

That's what your liver *should* be doing with saturated fats; making more cholesterol! The more the better! No mention is made of the far more harmful grain and seed oils (often labelled as *vegetable oils*) that are in packaged baked goods, with their inflammatory loads of omega-6 oils, and added sugars. The *only* oils mentioned are the two oils that have been shown by recent research to have a *beneficial* effect on atherosclerosis (ie: they *lessen* atherosclerotic plaque formation). If these oils are taken with fats sourced from cream cheese or tasty cheese, or even goose fat, all rich in vitamin K2, then the atherosclerotic plaque will be reversed.

What needs to be looked at a little closer is *why*, from all food sources, these particular beneficial tropical oils have deserved special mention. That's the subject of a whole book, which has probably already been written by several people with more expertise in the politics of nutrition than me.

What do shoe-laces and chromosomes have in common?

Meet Your Telomeres!

Melbourne University was the top-ranked university in Australia in 2015. One of the oldest Halls of Residence at Melbourne University is Janet Clarke Hall, where my scholarship-winning oldest daughter Annabel resided until recently. (As a very proud Dad I just had to put that in there!)

Janet Clarke Hall was where a Hobart-raised girl named Elizabeth Blackburn studied in the early 1960s. Elizabeth Blackburn won the 2009 Nobel Prize for Medicine and Physiology for her work on the chromosomal replenishment enzyme *telomerase*, and *telomeres*, the protein complexes which function much like the *aglets* on the ends of our shoelaces that prevent them from fraying.

The 'Shoelace-based Science of Chromosomal Breakdown'

Telomeres apparently unwind and shorten with aging and high *stress* levels. The genetic material at the ends of the chromosomes is replenished by the enzyme *telomerase.* Low telomerase levels are associated with 6 markers for heart disease.

How we eat, move, think and feel can either help keep our cells healthy or put them into early retirement, according to a growing body of research cited in Blackburn's new book, co-authored by Elissa Eppel, PhD, *The Telomere Effect: A Revolutionary Approach to Living Younger, Healthier, Longer*.

According to an online article by Adriana Barton of The Globe and Mail, Toronto, there are five main recommendations to draw from Professor Blackburn's research.

1.Telomeres Respond to two types of *Aerobic* Exercise

In a German study published in 2015, resistance exercise, such as weightlifting, had little effect on telomerase, the enzyme that replenishes telomeres. This is a different outcome to the production of BDNF, which is increased with resistance-training as well as aerobic exercise like walking, cycling or running.

But over a six-month period, two forms of 'aerobic' exercise increased telomerase activity twofold. One was moderate exercise, such as light jogging or fast walking, performed three times a week for at least 45 minutes. The other telomerase-friendly workout, also performed three times a week, was high-intensity interval training, consisting of a 10-minute warm-up, four alternating intervals of fast and easy running (at three minutes each) and a 10-minute cooldown.

(*Speaking as a long-time endurance athlete and coach, I would go for the low-intensity type first, as high-intensity intervals that last 3 minutes can take your system into moderate acidosis and immune-suppression if done too regularly. However, if you're fit and well, and still want to protect your telomeres, alternate the higher-intensity exercise with moderate aerobic recovery days, or do the 'intervals' one day a week. This is sufficient to promote the desired physiological response without 'frying' your aerobic system.)

2.Excess Sugar means ANY simple sugar.

Excess sugar consumption shortens lives.

In a 2014 study of 5,000 Americans, people who drank 590 millilitres of sugary pop a day had the equivalent of 4.6 extra years of biological aging (as measured by telomere shortness) compared with those who did not. Americans who drank 237 milliliters of pop daily had telomeres the equivalent of two years older. The link remained even after researchers ruled out other factors, including diet, smoking, income, age and body-mass index.

3.Telomeres thrive on fish, seaweeds and flaxseed oil

All are sources of omega-3 essential fatty acids, linked to longer telomeres. In a 2010 study, researchers from the University of California, San Francisco, tested blood levels of omega-3s in 608 middle-aged patients with heart disease. The higher their blood levels of omega-3s, the less their telomeres shortened over the next five years. Of those who had telomere shortening, 39 per cent died in the next four years, according to a follow-up report. Of those whose telomeres appeared to have lengthened, 12 per cent died – a significant difference considering that all participants had heart disease.

(*If you're a male wanting to increase your natural free testosterone levels, obtain your omega-3 fatty acids from fish oil, and avoid flax seed oil. Flax seed oil contains an excessive amount of lignans, plant compounds which annihilate free testosterone and can be considered a way to *biologically castrate* a male).

If you’re worried about possible mercury contamination in fish oil sourced from larger fish higher up in the food chain, this can be achieved by using oil sourced from marine creatures at the bottom of the food chain: krill, or shrimp, which are not only the main source of nutrition for the blue whale, but also one of nature’s richest sources of omega-3 fatty acids, with the benefit of having high levels of the powerful neuro-protective antioxidant and carotenoid, astaxanthin, which gives krill oil its distinctive dark purple-red colour.

4.Depression and anxiety deplete telomeres

In a 2015 study of nearly 12,000 Chinese women, depressed women had significantly shorter telomeres than women who weren't depressed. Chronic depression appears to be the most harmful. In a 2014 study of nearly 3,000 Dutch people, researchers found that the longer and more severe the depression, the shorter the telomeres.

Anxiety, pessimism, hostility, mind wandering and rumination have also been linked to shorter telomeres. Fortunately, practices known to help us break these mental habits may help lengthen telomeres. (My personal way of countering those tendencies is to recite the ‘Fruits of the Spirit’ and The 23rd Psalm each evening on my peaceful 35-minute bush-walk).

In a 2012 study, researchers divided 64 people with chronic-fatigue syndrome into a control group and a group who learned *qigong*, a Chinese practice that emphasizes meditative movements and breathing. After four months, participants who practiced *qigong* had significantly greater increases in telomerase activity, and reductions in fatigue, than those not doing *qigong*.

5.No Yo-Yo Dieting and No Sugar

Repeated weight loss and weight gain (yo-yo dieting) appears to shorten telomeres. Instead of focusing on the numbers on the scale, the authors write, we should take steps to reduce excess belly fat (as opposed to fat on the hips and thighs) and improve our metabolic health.

One way is to lower our sugar intake.

While the war on sugar is nothing new, Telomere science quantifies the degree to which excess sugar consumption shortens lives.

In a 2014 study of 5,000 Americans, people who drank 590 millilitres of sugary pop a day (1 standard bottle) had the equivalent of 4.6 extra years of biological aging (as measured by telomere shortness) compared with those who did not. Americans who drank 237 millilitres of pop daily had telomeres the equivalent of two years older. The link remained even after researchers ruled out other factors, including diet, smoking, income, age and body-mass index.

Further Suggested Reading

These books are thoroughly recommended for everyone who wishes to improve with brain health and nutrition.

Brain Maker–The Power of Gut Microbes to Heal and Protect Your Brain–for Life. Dr David Perlmutter with Kristin Loberg

10% Human: How Your Body's Microbes Hold the Key to Health and Happiness . Alanna Cohen (This is a comprehensive book from a microbiologist who has been studying the microbiome of the gut for decades)

Excitotoxins The Taste that Kills Russell L. Blaylock, M.D.

Stop Alzheimer's Now! Bruce Fife, N.D. Foreword by Russell L. Blaylock, M.D. (A great, highly referenced book that is extremely high on detail; it's a 'classic' in the new field of metabolic nutrition)

The BIG FAT SURPRISE Why Butter, Meat & Cheese Belong in a Healthy Diet Nina Teicholz (Nine years of research by a journalist and New York Times best-selling author)

Primal Endurance Mark Sisson and Brad Kearns. (When 60 year olds look like these guys, sit up and take notice!)

21 Day Total Body Transformation Mark Sisson (This book explains how to transition your diet into a primal, ketogenic diet in three weeks).

Keto Clarity Your Definitive Guide to the Benefits of a Low-Carb, High-Fat Diet Jimmy Moore with Eric C.Westman, MD, and a panel of 22 experts

Presence: Bringing Your Boldest Self to Your Biggest Challenges by Amy Cuddy

Life on Purpose: How Living for What Matters Most Changes Everything by Victor Strecher

Man's Search for Meaning by Victor E Frankl

The Telomere Effect: A Revolutionary Approach to Living Younger, Healthier, Longer Elissa Eppel PhD, Elizabeth Blackburn PhD.